

Project Information - Dataset 1

RGB-D Apple-Tree Reconstruction and Orchard Mapping [Dataset Available]

General Problem

Develop a computational method that converts and aligns RGB-D observations into a useful three-dimensional representation of apple trees or orchard rows. The broader engineering goal is to support perception and path planning for agricultural robots.

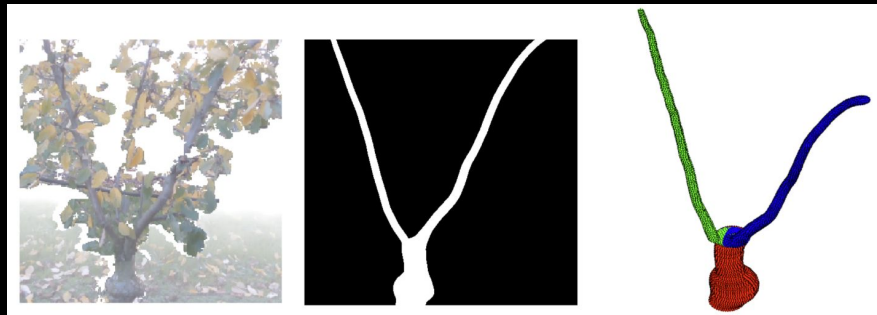
Aim to define a narrower and achievable problem within this context. Possible directions could include:

- reconstructing individual trees;
- extracting a three-dimensional tree skeleton;
- aligning observations captured from different viewpoints;
- matching trees across seasons;
- localising apple on summer trees; or
- evaluating whether the reconstruction is suitable for robot path planning.

Main Inputs Provided

- RGB images containing colour and visual information.
- Depth observations providing estimated distance from the camera.
- Observations collected across multiple trees and seasons.
- Orchard row and tree-column identifiers, where available.
- GPS or other position information, maybe.
- Additional seasonal datasets that may contain incomplete position metadata.

Not every season or observation is expected to contain every metadata field. Handling incomplete or unmatched data may form part of the engineering problem.



Project Information - Dataset 2

Monash Smart Infrastructure Occupancy and Environmental Data

General Problem

Develop a computational method that uses time-stamped building sensor data to understand, estimate or predict occupancy and environmental behaviour.
Aim to define a specific and achievable question. Possible directions could include:

- predicting room occupancy;
- reconstructing missing sensor measurements;
- identifying daily or weekly operating patterns;
- detecting unusual environmental conditions;
- comparing rooms or operating periods;
- investigating relationships between occupancy and environmental variables; or
- evaluating whether the available measurements support decisions about comfort or energy use.

Main Inputs Provided

Subject to the final data extract:

- time-stamped occupancy information;
- available environmental sensor measurements;
- room, zone or building identifiers;
- sensor or device identifiers;
- measurement units and sensor metadata;
- repeated measurements over time; and
- missing, irregular or potentially unreliable observations that require auditing.



Project Information - Dataset 3

Automated Detection of Pork Rasher Packaging Errors and Meat Quality [Dataset Available]

General Problem

Develop a computer-vision method that analyses images of packaged pork rashers to detect and localise packaging or product-placement defects.

The broader engineering goal is to support automated quality inspection in food-packaging operations, reducing reliance on manual inspection and identifying defective products before distribution.

Aim to define a narrower and achievable problem within this context. Possible directions could include:

- detecting and localising packaging defects;
- distinguishing between loose meat, twisted meat, unsealed packaging and wrinkles;
- determining whether a tray should pass or fail inspection;
- counting rashers or estimating their position and orientation within a tray;
- measuring the severity or extent of a detected defect;
- comparing the performance of different object-detection models;
- investigating errors caused by reflections, occlusion or overlapping meat; or
- evaluating whether the method is sufficiently accurate and fast for real-time production-line inspection.

Main Inputs Provided

- RGB images of pork rasher trays and packaging;
- object-detection bounding-box annotations;
- approximately 1,955 original project images;
- a generated Version 4 dataset containing 3,549 images after augmentation;
- predefined training, validation and test splits;
- images auto-oriented and resized to 720 × 540 pixels; and
- augmented training images with variations in saturation and exposure.

